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Educational History

May 2018 PhD (Chemistry), University of Notre Dame, South Bend, IN
Thesis Advisor: Bradley D. Smith, Emil T. Hofman Professor of Chemistry
Thesis Title: “Molecular Recognition Using Tetralactam Macrocycle and Development of Synthavidin Technology”
Jun 2013 BS (Chemistry), Shandong University, Shandong, China
Research Advisor: Prof. Aiyu Hao

Employment History

Aug 2026 – Now Associate Professor, Department of Chemistry, University of South Florida, Tampa, FL
Aug 2021 – Jul 2026 Assistant Professor, Department of Chemistry, University of South Florida, Tampa, FL
Mar. 2022 – Present Affiliate Faculty, Department of Chemical, Biological and Materials Engineering, Tampa, University of South Florida, Tampa, FL
Aug 2018 – Jul 2021 Postdoctoral Fellow, Department of Chemistry, Northwestern, Evanston, IL
Research Advisor: Sir Fraser Stoddart, Board of Trustee Professor of Chemistry, Nobel Laureate in Chemistry 2016

Awards

2024 NSF Early Career Award
2023 Journal of Materials Chemistry Emerging Investigator
2018 Dow Chemical Company Outstanding Graduate Student Award (University of Notre Dame, IN)

Research Area:

With a focus on molecular design and synthesis, my research group encompasses supramolecular chemistry, organic synthesis, physical chemistry, polymer chemistry, and advanced materials. A central theme of our research is focused on molecular recognition, where new physical, chemical, optical, and mechanical properties emerge from noncovalent interactions. Our research aims to tackle critical societal challenges in energy, health, and environmental sciences. The major goals in my group’s research are:

1. Understanding Hydrogen Bonding in Water:

One of our primary objectives is to unravel the fundamental principles underlying hydrogen bonding in water. This ubiquitous yet enigmatic interaction plays a pivotal role in various chemical and biological processes, holding the key to understanding life's processes and developing transformative technologies in the biomedical field.

2. Developing Molecular Containers

We strive to develop effective synthetic strategies for the creation of molecular containers. These containers possess the remarkable ability to encapsulate specific molecules, paving the way for innovative applications in sensing, catalysis, and separation.

3. Synthetic Lectins for Diabetes Management:

Our group is dedicated to the development of synthetic lectins, which have the potential to revolutionize diabetes management. We aim to create glucose-responsive peptides (insulins and glucagon) that could provide a groundbreaking approach to diabetes treatment and management.

4. Selective Bimolecular Sensing:

Another key objective is the development of highly selective molecular containers for continuous biomolecular sensing. This investigation has a far-reaching impact on the fields of disease diagnostics and management.

5. Micro Pollutant Sequestration:

Our research addresses the critical issue of micro-pollutant sequestration, aiming to develop efficient methods based on molecular recognition technology for the removal of pollutants from water sources, thereby safeguarding the environment and human health.

6. Sustainable Separation of Critical Minerals:

We are actively engaged in devising sustainable approaches for the separation of critical minerals. This research, based on weak interactions, holds great promise in ensuring a stable supply of essential materials for emerging technologies.

Publications: 49 / Total citations: 2192 / h-index: 26 / i10-index: 44

(*Denotes Corresponding Authorship, Undergraduate Authors are Underlined):

Peer-Reviewed Publications

51. Niu, S.; Jiang, Y.; Lewis, D. W.; Owoseeni, O. E.; Hassan, M. M.; Arulsamy, N.; Stephenson, T.; Anderson, K.; Yang, Q.; Yao, Y.; Mariscal, A.; Liu, W.; Duan, C.; Li, P.; Zhang, X.; Goroncy, A. K.; Li, X.. Mechanically Planar Chiral Molecules by a SuFEx-Auxiliary Approach. *Angew. Chem. Int. Ed.* **2026**, 65, e25515.
50. Liu, D.; Victoria, A.; Conces, J.; Mampilly, S.; Mariscal, A.; Cui, Y.; Wojtas, L.; Cai, J.; **Liu, W.*** Shape-Shifting Tetralactam Macrocycles: Protonation-Activated Convergent Hydrogen Bonding. *Chem. Eur. J.* **2026**, 32, e03496
49. Zhai, C.; Xu, C.; Cui, Y.; Wojtas, L.; Cai, J.; **Liu, W.*** A Synthetic Lectin for Glucuronate. *ACS Cent. Sci.* **2025**, 11, 1753–1761. (Supplementary cover)
48. Zhou, P.; Cheng, K.; Qu, K.; Wang, L.; Hu, C.; **Liu, W.*** Chen, H. An electric molecular Faraday cage. *J. Am. Chem. Soc.* **2025**, 147, 19272–19281.
47. Huang, B.; Li, S.; Pan, C.; Li, F.; Wojtas, L.; Qiao, Q.; Tran, T. H.; Calcul, L.; **Liu, W.**; Ke, C.; Cai, J. Proline-Based Tripodal Cages with Guest-Adaptive Features for Capturing Hydrophilic and Amphiphilic Fluoride Substances. *Nat. Commun.* **2025**, 16, 3226.
46. Mariscal, A.; Sagal, L.; Doan, C.; Zhai, C.; Liu, D.; Wojtas, L.; **Liu, W.*** Sulfate Recognition in Water via Charge-Assisted Hydrogen Bonding. *Chem. Eur. J.* **2025**, 31, e202501400
45. Zhai, C.; Mariscal, A.; **Liu, W.***; Molecular Recognition in Water by Synthetic Hydrogen-Bonding Receptors. *Trends Chem.*, **2025**, 7, 70–84. (Invited, Front Cover)
44. Wu, H.; Wang, Y.; Dordevic, L.; Kundu, P.; Bhunia, S.; Chen, A. X.-Y.; Feng, L.; Shen, D.; **Liu, W.**; Zhang, L.; Song, B.; Wu, G.; Liu, B.-T.; Yang, M. Y.; Stern, C. L.; Stupp, S. I.; Goddard III, W.; Hu, W.; Stoddart, J. F. Dynamic Supramolecular Snub Cubes. *Nature* **2025**, 637, 347–353.
43. Zhai, C.; Zulueta, E. C.; Mariscal, A.; Xu, C.; Cui, Y.; Wang, X.; Wu, H.; Doan, C.; Wojtas, L.; Zhang, H.; Cai, J.; Ye, L.; Wang, K.; **Liu, W.*** From Small Changes to Big Gains: Pyridinium-Based Tetralactam Macrocyclic for Enhanced Sugar Recognition in Water. *Chem. Sci.* **2024**, 15, 19588–19598.
42. Xu, C.; Tran, Q.; Liu, D.; Zhai, C.; Wojtas, L.; **Liu, W.*** Charge-Assisted Hydrogen Bonding in A Bicyclic Amide Cage: An Effective Approach to Anion Recognition and Catalysis in Water. *Chem. Sci.* **2024**, 15, 16040–16049. (Back Cover)
41. Feng, Y.; Zhao, X.; Appleton, D. A.; Han, H.; Young, R. M.; **Liu, W.**; Lee, C. K.; Li, W.; Liu, B.-T.; Wu, Y.; Tang, C.; Chen, A. X.-Y.; Stern, C. L.; Kim, D. J.; Wasielewski, M. R.; Qiu, Y.; Stoddart, J. F. Chalcogenoviologen Enhanced Host-Guest Recognition. *CCS Chem.* **2024**, 6, 2679–2681.
40. Tang, C.; Han, H.; Zhang, R.; De Moraes, L. S.; Qi, Y.; Wu, G.; Jones, C. G.; Rodriguez, I. H.; Jiao, Y.; **Liu, W.**; Li, X.; Chen, H.; Bancroft, L.; Zhao, X.; Stern, C. L.; Guo, Q.-H.; Krzyaniak, M. D.; Wasielewski, M. R.; Nelson, H. M.; Li, P.; Stoddart, J. F. A Geometrically Flexible Three-Dimensional Nanocarbon. *J. Am. Chem. Soc.* **2024**, 146 (29), 20158–20167.

39. Xue, S.; Xu, W.; Wang, L.; Wang, X.; Duan, Q.; Calcul, L.; Wang, S.; **Liu, W.**; Sun, X.; Lu, L.; Jiang, S.; Cai, J. An HR2-Mimicking Sulfonyl- γ -AApeptide Is a Potent Pan-Coronavirus Fusion Inhibitor with Strong Blood–Brain Barrier Permeability, Long Half-Life, and Promising Oral Bioavailability. *ACS Cent. Sci.* **2023**, *9* (5), 1046–1058.
38. Xu, C.; Tran, Q.; Wojtas, L.; **Liu, W.***. Harnessing Ion–Dipole Interactions: A Simple and Effective Approach to High-Performance Lithium Receptors. *J. Mater. Chem. A* **2023**, *11* (23), 12214–12222.
37. Chen, H.; Roy, I.; Myong, M. S.; Seale, J. S. W.; Cai, K.; Jiao, Y.; **Liu, W.**; Song, B.; Zhang, L.; Zhao, X.; Feng, Y.; Liu, F.; Young, R. M.; Wasielewski, M. R.; Stoddart, J. F. Triplet–Triplet Annihilation Upconversion in a Porphyrinic Molecular Container. *J. Am. Chem. Soc.* **2023**, *145* (18), 10061–10070.
36. Zhai, C.; Xu, C.; Cui, Y.; Wojtas, L.; **Liu, W.***. Dynamic Approach to Synthetic Lectin for Glucose with Boosted Binding Affinity through C–H Hydrogen Bonds. *Chem. Eur. J.* **2023**, *29* (32), e202300524.
35. Wu, H.; Wang, Y.; Tang, C.; Jones, L. O.; Song, B.; Chen, X. Y.; Zhang, L.; Wu, Y.; Stern, C. L.; Schatz, G. C.; **Liu, W.***; Stoddart, J. F. High-Efficiency Gold Recovery by Additive-Induced Supramolecular Polymerization of β -Cyclodextrin. *Nat. Commun.* **2023**, *14* (1), 1284.
34. Garci, A.; David, A. H. G.; Le Bras, L.; Ovalle, M.; Abid, S.; Young, R. M.; **Liu, W.**; Azad, C. S.; Brown, P. J.; Wasielewski, M. R.; Stoddart, J. F. Thermally Controlled Exciplex Fluorescence in a Dynamic Homo[2]Catenane. *J. Am. Chem. Soc.* **2022**, *144* (51), 23551–23559.

Publication Prior to Independent Career

33. Garci, A.; Weber, J. A.; Young, R. M.; Kazem-Rostami, M.; Ovalle, M.; Beldjoudi, Y.; Atilgan, A.; Bae, Y. J.; **Liu, W.**; Jones, L. O.; Stern, C. L.; Schatz, G. C.; Farha, O. K.; Wasielewski, M. R.; Fraser Stoddart, J. Mechanically Interlocked Pyrene-Based Photocatalysts. *Nat. Cat.* **2022**, *5* (6), 524–533.
32. Wu, Y.; Guo, Q. H.; Qiu, Y.; Weber, J. A.; Young, R. M.; Bancroft, L.; Jiao, Y.; Chen, H.; Song, B.; **Liu, W.**; Feng, Y.; Zhao, X.; Li, X.; Zhang, L.; Chen, X. Y.; Li, H.; Wasielewski, M. R.; Stoddart, J. F. Syntheses of Three-Dimensional Catenanes under Kinetic Control. *Proc. Natl. Acad. Sci. USA* **2022**, *119* (12), 1–6.
31. **Liu, W.**; Tan, Y.; Jones, L. O.; Song, B.; Guo, Q. H.; Zhang, L.; Qiu, Y.; Feng, Y.; Chen, X. Y.; Schatz, G. C.; Stoddart, J. F. PCage: Fluorescent Molecular Temples for Binding Sugars in Water. *J. Am. Chem. Soc.* **2021**, *143* (38), 15688–15700.
30. **Liu, W.**; Das, P. J.; Colquhoun, H. M.; Stoddart, J. F. Whither Second-Sphere Coordination? *CCS Chem.* **2022**, *4* (3), 755–784.
29. Wu, H.; Wang, Y.; Song, B.; Wang, H.-J.; Zhou, J.; Sun, Y.; Jones, L. O.; **Liu, W.**; Zhang, L.; Zhang, X.; Cai, K.; Chen, X.-Y.; Stern, C. L.; Wei, J.; Farha, O. K.; Anna, J. M.; Schatz, G. C.; Liu, Y.; Fraser Stoddart, J. A Contorted Nanographene Shelter. *Nat. Commun.* **2021**, *12* (1), 5191.
28. Wu, H.; Wang, Y.; Jones, L. O.; **Liu, W.**; Zhang, L.; Song, B.; Chen, X. Y.; Stern, C. L.; Schatz, G. C.; Stoddart, J. F. Selective Separation of Hexachloroplatinate (IV) Dianions Based on Exo-Binding with Cucurbit[6]uril. *Angew. Chem. Int. Ed.* **2021**, *60* (32), 17587–17594.
27. **Liu, W.**; Stoddart, J. F. Emergent Behavior in Nanoconfined Molecular Containers. *Chem* **2021**, *7* (4), 919–947.
26. **Liu, W.**; Jones, L. O.; Wu, H.; Stern, C. L.; Sponenburgh, R. A.; Schatz, G. C.; Stoddart, J. F. Supramolecular Gold Stripping from Activated Carbon Using α -Cyclodextrin. *J. Am. Chem. Soc.* **2021**, *143* (4), 1984–1992.

25. Anamimoghadam, O.; Jones, L. O.; Cooper, J. A.; Beldjoudi, Y.; Nguyen, M. T.; **Liu, W.**; Krzyaniak, M. D.; Pezzato, C.; Stern, C. L.; Patel, H. A.; Wasielewski, M. R.; Schatz, G. C.; Stoddart, J. F. Discrete Open-Shell Tris(Bipyridinium Radical Cationic) Inclusion Complexes in the Solid State. *J. Am. Chem. Soc.* **2021**, *143* (1), 163–175.
24. Cai, K.; Cui, B.; Song, B.; Wang, H.; Qiu, Y.; Jones, L. O.; **Liu, W.**; Shi, Y.; Vemuri, S.; Shen, D.; Jiao, T.; Zhang, L.; Wu, H.; Chen, H.; Jiao, Y.; Wang, Y.; Stern, C. L.; Li, H.; Schatz, G. C.; Li, X.; Stoddart, J. F. Radical Cyclic [3]Daisy Chains. *Chem* **2021**, *7* (1), 174–189.
23. Shaffer, C. C.; **Liu, W.**; Oliver, A. G.; Smith, B. D. Supramolecular Paradigm for Capture and Co-Precipitation of Gold(III) Coordination Complexes. *Chem. Eur. J.* **2021**, *27* (2), 751–757.
22. Wu, H.; Wang, Y.; Jones, L. O.; **Liu, W.**; Song, B.; Cui, Y.; Cai, K.; Zhang, L.; Shen, D.; Chen, X.-Y.; Jiao, Y.; Stern, C. L.; Li, X.; Schatz, G. C.; Stoddart, J. F. Ring-in-Ring(s) Complexes Exhibiting Tunable Multicolor Photoluminescence. *J. Am. Chem. Soc.* **2020**, *142* (39), 16849–16860.
21. Qiu, Y.; Song, B.; Pezzato, C.; Shen, D.; **Liu, W.**; Zhang, L.; Feng, Y.; Guo, Q. H.; Cai, K.; Li, W.; Chen, H.; Nguyen, M. T.; Shi, Y.; Cheng, C.; Dean Astumian, R.; Li, X.; Fraser Stoddart, J. A Precise Polyrotaxane Synthesizer. *Science* **2020**, *368* (6496), 1247–1253.
20. **Liu, W.**; Stern, C. L.; Stoddart, J. F. Suit[4]Ane. *J. Am. Chem. Soc.* **2020**, *142* (23), 10273–10278.
19. **Liu, W.**; Lin, C.; Weber, J. A.; Stern, C. L.; Young, R. M.; Wasielewski, M. R.; Stoddart, J. F. Cyclophane-Sustained Ultrastable Porphyrins. *J. Am. Chem. Soc.* **2020**, *142* (19), 8938–8945.
18. **Liu, W.**; Bobbala, S.; Stern, C. L.; Hornick, J. E.; Liu, Y.; Enciso, A. E.; Scott, E. A.; Fraser Stoddart, J. XCage: A Tricyclic Octacationic Receptor for Perylene Diimide with Picomolar Affinity in Water. *J. Am. Chem. Soc.* **2020**, *142* (6), 3165–3173.
17. **Liu, W.**; Oliver, A. G.; Smith, B. D. Stabilization and Extraction of Fluoride Anion Using a Tetralactam Receptor. *J. Org. Chem.* **2019**, *84* (7), 4050–4057.
16. McGarraugh, H. H.; **Liu, W.**; Matthews, B. P.; Smith, B. D. Croconaine Rotaxane Dye with 984 nm Absorption: Wavelength-Selective Photothermal Heating. *Eur. J. Org. Chem.* **2019**, *2019* (21), 3489–3494.
15. **Liu, W.**; McGarraugh, H. H.; Smith, B. D. Fluorescent Thienothiophene-Containing Squaraine Dyes and Threaded Supramolecular Complexes with Tunable Wavelengths between 600–800 nm. *Molecules* **2018**, *23* (9), 2229.
14. **Liu, W.**; Oliver, A. G.; Smith, B. D. Macrocyclic Receptor for Precious Gold, Platinum, or Palladium Coordination Complexes. *J. Am. Chem. Soc.* **2018**, *140* (22), 6810–6813.
13. **Liu, W.**; Johnson, A.; Smith, B. D. Guest Back-Folding: A Molecular Design Strategy That Produces a Deep-Red Fluorescent Host/Guest Pair with Picomolar Affinity in Water. *J. Am. Chem. Soc.* **2018**, *140* (9), 3361–3370.
12. Shaw, S. K.; **Liu, W.**; Gómez Durán, C. F. A.; Schreiber, C. L.; Betancourt Mendiola, M. de L.; Zhai, C.; Roland, F. M.; Padanilam, S. J.; Smith, B. D. Non-Covalently Pre-Assembled High-Performance Near-Infrared Fluorescent Molecular Probes for Cancer Imaging. *Chem. Eur. J.* **2018**, *24* (52), 13821–13829.
11. **Liu, W.**; Gómez-Durán, C. F. A.; Smith, B. D. Fluorescent Neuraminidase Assay Based on Supramolecular Dye Capture after Enzymatic Cleavage. *J. Am. Chem. Soc.* **2017**, *139* (18), 6390–6395.
10. **Liu, W.**; Samanta, S. K.; Smith, B. D.; Isaacs, L. Synthetic Mimics of Biotin/(Strept)Avidin. *Chem. Soc. Rev.* **2017**, *46* (9), 2391–2403.

9. Shaw, S. K.; **Liu, W.**; Brennan, S. P.; de Lourdes Betancourt-Mendiola, M.; Smith, B. D. Non-Covalent Assembly Method That Simultaneously Endows a Liposome Surface with Targeting Ligands, Protective PEG Chains, and Deep-Red Fluorescence Reporter Groups. *Chem. Eur. J.* **2017**, *23* (51), 12646–12654.
8. Gómez-Durán, C. F. A.; **Liu, W.**; Betancourt-Mendiola, M. D. L.; Smith, B. D. Structural Control of Kinetics for Macrocycle Threading by Fluorescent Squaraine Dye in Water. *J. Org. Chem.* **2017**, *82* (16), 8334–8341.
7. Hu, D.; Jin, S.; Shi, Y.; Wang, X.; Graff, R. W.; **Liu, W.**; Zhu, M.; Gao, H. Preparation of Hyperstar Polymers with Encapsulated Au₂₅(SR)₁₈ Clusters as Recyclable Catalysts for Nitrophenol Reduction. *Nanoscale* **2017**, *9* (10), 3629–3636.
6. **Liu, W.**; Peck, E. M.; Smith, B. D. High Affinity Macrocycle Threading by a Near-Infrared Croconaine Dye with Flanking Polymer Chains. *J. Phys. Chem. B* **2016**, *120* (5), 995–1001.
5. **Liu, W.**; Peck, E. M.; Hendzel, K. D.; Smith, B. D. Sensitive Structural Control of Macrocycle Threading by a Fluorescent Squaraine Dye Flanked by Polymer Chains. *Org. Lett.* **2015**, *17* (21), 5268–5271.
4. Peck, E. M.; **Liu, W.**; Spence, G. T.; Shaw, S. K.; Davis, A. P.; Destecroix, H.; Smith, B. D. Rapid Macrocycle Threading by a Fluorescent Dye-Polymer Conjugate in Water with Nanomolar Affinity. *J. Am. Chem. Soc.* **2015**, *137* (27), 8668–8671.
3. Li, Z.; **Liu, W.**; Hao, A. Gel-Sol-Gel' Evolution Triggered by Formic Acid. *Colloids Surf. A: Physicochem. Eng. Aspects* **2014**, *451*, 25–32.
2. Hou, Y.; Li, S.; Sun, T.; Yang, J.; Xing, P.; **Liu, W.**; Hao, A. Organogels Based on β -Cyclodextrin System with Molecular Recognition Property. *J Incl Phenom Macrocycl Chem* **2014**, *80* (3–4), 217–224.
1. **Liu, W.**; Xing, P.; Xin, F.; Hou, Y.; Sun, T.; Hao, J.; Hao, A. Novel Double Phase Transforming Organogel Based on β -Cyclodextrin in 1,2-Propylene Glycol. *J. Phys. Chem. B* **2012**, *116* (43), 13106–13113.

Patents (Both US and International)

7. Liu, W. Pyridinium-Based Cyclophanes as Synthetic Lectins and Their Use for Glucose Recognition and Sensing in Aqueous Solutions. International Patent filed: 11001-205WO1
6. Liu, W. High-efficiency gold recovery by additive-induced supramolecular polymerization of β -cyclodextrin. Under IP disclosure.
5. Stoddart, J.F.; Liu, W.; Tan, Y. Pyrene-Based Cyclophanes as Synthetic Lectins and Their Use for Glucose Recognition and Sensing in Aqueous Solutions. Under IP disclosure.
4. Stoddart, J. F.; Liu, W.; Supramolecular Gold Stripping from Activated Carbon Using α -Cyclodextrin. US Patent Application No. 18/258,431
3. Stoddart, J. F.; Liu, W.; Cyclophane-Sustained High-Performance Porphyrins. US Patent 12,084,579.
2. Stoddart, J. F.; Liu, W.; Tricyclic Octacationic Cyclophane and Its Use in Complexation with Perlene Diimide Dyes. Patent. US Patent Application No. 17/445,053.
1. Xing, P.; Hao, A.; Liu, W.; Non-toxic and Stable Small Molecular Organic Gel and Preparation Method Thereof. China Patent No. CN102,627,790 B

Lectures

(Future) Invited and Contributed Lectures (In a department of chemistry unless otherwise noted)

32. Liu, W. Taming Hydrogen Bonding in Water. 2026 Fall ACS meeting Stoddart Symposium, Chicago, IL, Aug. 25, 2026 (Invited)
31. Liu, W. Discovering synthetic molecular receptors for sustainable separation of lithium. 2026 Fall ACS meeting, Chicago, IL, Aug. 24, 2026 (Contributed)

30. Liu, W. Taming Hydrogen Bonding in Water. International Symposium on Macrocyclic and Supramolecular Chemistry (ISMSC) 2026, Bordeaux, France, Jul. 10, 2026 (Contributed).
29. Liu, W. Taming Hydrogen Bonding in Water. Washington University in St. Louis, St. Louis, MO, Apr. 9 2026 (Invited)
28. Liu, W. Anion Recognition in Water by Hydrogen Bonding Receptors. Anion Recognition Chemistry symposium PacifiChem 2025, Honolulu, HI, Dec. 15, 2025 (Invited).
27. Liu, W. Taming Hydrogen Bonding in Water. University of Pennsylvania, Philadelphia, PA, Nov. 17, 2025. (Invited)
26. Liu, W. Taming Hydrogen Bonding in Water. University of Oregon, Eugene, OR, Nov. 12, 2025. (Invited)
25. Liu, W. Taming Hydrogen Bonding in Water. University of Colorado, Boulder, CO, Nov. 10, 2025 (Invited)
24. Liu, W. Taming Hydrogen Bonding in Water. Georgetown University, Washington D.C., Nov. 7, 2025 (Invited)
23. Liu, W. Taming Hydrogen Bonding in Water. University of Maryland, College Park, MD, Nov. 6, 2025. (Invited)
22. Liu, W. Taming Hydrogen Bonding in Water. University of California, Riverside, CA, Oct. 24, 2025. (Invited)
21. Liu, W. Taming Hydrogen Bonding in Water. University of Notre Dame, South Bend, IN, Oct. 9, 2025. (Invited)
20. Liu, W. Taming Hydrogen Bonding in Water. University of Central Florida, Orlando, FL, Sep. 26, 2025 (Invited)
19. Liu, W. Taming Hydrogen Bonding in Water. Indiana University Bloomington, IN, Sep. 10, 2025 (Invited)
18. Liu, W. Taming Hydrogen Bonding in Water. Purdue University, West Lafayette, IN, Sep. 9, 2025 (Invited)
17. Liu, W. Taming Hydrogen Bonding in Water. Florida State University, Tallahassee, FL, Sep. 4, 2025 (Invited)
16. Liu, W. Taming Hydrogen Bonding in Water. University of Florida, Gainesville, FL, Sep. 1, 2025 (Invited).
15. Liu, W. Taming Hydrogen Bonding in Water. 5th Aqueous Supramolecular Chemistry Workshop, Montana State University, MT, Jul. 21, 2025 (Invited).
14. Liu, W. Taming Hydrogen Bonding in Water. University of Osaka, Osaka, Japan, Jun. 4, 2025 (Invited).
13. Liu, W. Taming Hydrogen Bonding in Water. International Symposium on Macrocyclic and Supramolecular Chemistry (ISMSC) 2025, Kyoto, Japan, May, 26, 2025 (Contributed).
12. Liu, W. Taming Hydrogen Bonding in Water. Dartmouth College, Hanover, NH, May 13, 2025 (Invited)
11. Liu, W. NSF Career Workshop panelist, University of South Florida, Tampa, Nov. 15, 2024.
10. Liu, W. Taming Hydrogen Bonding in Water. University of Wyoming, Laramie, WY, Oct. 25, 2024
9. Liu, W. Molecular Recognition of Hydrophilic Substates in Water Through Hydrogen Bonding Receptors. International Symposium on Macrocyclic and Supramolecular Chemistry (ISMSC) 2024, Hangzhou, China, May, 06, 2024.
8. Liu, W. Dynamic Approaches to Synthetic Receptors for Molecular Recognition of Hydrophilic Substrates in Water. Chinese American Chemistry & Chemical Biology Professor Associations, Tampa, FL, Dec. 20, 2023
7. Liu, W. Dynamic Approaches to Synthetic Receptors for Molecular Recognition of Hydrophilic Substrates in Water. North American Supramolecular Chemistry meeting, New Orleans, LA, USA, Dec. 18, 2023
6. Liu, W. Lessons from Molecular Switches and Machines. iCANX Online forum, Oct. 24, 2023
5. Liu, W. Tailor-Made Molecular Containers for Molecular Recognition at Atomic Resolution. University of West Florida, Department of Chemistry, Pensacola, FL, USA, Aug. 20, 2023
4. Liu, W. Tailor-Made Molecular Containers for Molecular Recognition at Atomic Resolution. Southeastern University, Department of Chemistry, Lakeland, FL, USA, Sep. 22, 2022
3. Liu, W. Tailor-Made Molecular Containers for Molecular Recognition at Atomic Resolution. University of South Florida, Department of Chemical Biological and Materials Engineering, Tampa, FL, USA, Mar. 02, 2022
2. Liu, W. Tailor-Made Molecular Containers for Molecular Recognition at Atomic Resolution. University of South Florida, Department of Chemistry, Tampa, FL, USA, Feb. 16, 2022
1. Liu, W.; Gómez-Durán, C. F. A.; Smith, B. D. Fluorescent Enzyme Assay Based on Pseudorotaxane Formation. Oral Presentation Delivered at 252th ACS National Meeting, Philadelphia, PA, Aug. 2016.

Contributed Posters

1. Liu, W.; Gómez-Durán, C. F. A.; Smith, B. D. Fluorescent Neuraminidase Assay Based on Supramolecular Dye Capture after Enzymatic Cleavage. Poster Presentation Delivered at 14th Annual Conference on Foundations of Nanoscience, Snowbird, Utah, April 2017.

Students Contributed Talks and Posters (Undergraduates are underlined)

8. Phan, T. Charge-assisted hydrogen bonding receptor as an effective approach to anion recognition and catalysis in water, Tampa, FL, USA, April 26, 2025
7. Maji, I. A Hydrogen Bonding Receptor for Dihydrogen Phosphate. Castle Conference, Tampa, FL, USA, April 26, 2025
6. Zhai, C. Pyridinium-based tetralactam macrocycle for enhanced sugar recognition in water. NASC 2024, New Orleans, USA, Dec. 17, 2024.
5. Xu, C. Charge-assisted hydrogen bonding in a bicyclic amide cage: an effective approach to anion recognition and catalysis in water. NASC 2024, New Orleans, USA, Dec. 17, 2024.
4. Mariscal, A. Charge-assisted anion recognition in water by a “saddle lactam” macrocycle. Florida Inorganic and Materials Symposium 2024, Gainesville, FL. USA. Nov. 8, 2024.
3. Sagal, L. Charge-Assisted Anion Recognition in Water. Castle Conference, Tampa, FL, USA, Mar. 2, 2024.
2. Colmegna, T. Pyrrole Based Molecular Cage, Castle Conference, Tampa, FL, USA, Mar. 2, 2024.
1. Xu, C.; Tran, Q. Harnessing ion-dipole interactions: a simple and effective approach to high-performance lithium receptors. Castle Conference, Tampa, FL, USA, Mar. 4, 2023.

Funding Awarded

CAREER: Taming Hydrogen Bonding in Water	
PI:	Wenqi Liu
Agency:	NSF-MSN, CHE-2337419
Type & Dates:	Federal grant 08/01/2024–07/31/2029
Funding:	\$599,879
Aim:	To investigate molecular recognition in water dictated by hydrogen bonding
Discovering Synthetic Molecular Receptors for Sustainable Separation of Lithium	
PI:	Wenqi Liu
Agency:	NSF-CBET, ENG-2513233
Type & Dates:	Federal grant , 07/01/2025–6/30/2028
Funding:	\$464,985
Aim:	To develop direction lithium extraction technology
Travel award	
PI:	Wenqi Liu
Agency:	Sponsored Research, USF
Type & Dates:	CAS-ORS Conference Presenter Support program 2026
Funding:	\$750
Aim:	To cover the ISMSC 2026 registration fee.
Travel award	
PI:	Wenqi Liu
Agency:	Sponsored Research, USF
Type & Dates:	CAS-ORS Conference Presenter Support program 2025
Funding:	\$750
Aim:	To cover the ISMSC 2025 registration fee.
Developing Supramolecular Adsorbent Materials for Direct Ammonium Extraction	
PI:	Wenqi Liu
Agency:	Sponsored Research, USF
Type & Dates:	Internal New Researcher Grant 05/01/2025–04/30/2026
Funding:	\$20,000
Aim:	To discover molecular receptors for ammonium

Macrocyclic Ureas for Selective Capture and Release of Lithium Ions	
PI:	Wenqi Liu
Agency:	Sponsored Research, USF
Type & Dates:	Internal New Researcher Grant 05/01/2022–04/30/2023
Funding:	\$10,000
Aim:	To discover molecular receptors for lithium ions.
Startup at the University of South Florida	
PI:	Wenqi Liu
Agency:	USF
Type & Dates:	Startup funding 08/07/2021–08/07/2026
Funding:	\$450,000
Aim:	To establish the research group at USF covering equipment purchase, postdoc salary and consumables

Funding Pending

Synthetic Mimic of Phosphate Binding Protein	
PI:	Wenqi Liu
Agency:	NIH-MIRA-R35
Type	Federal grant
Expected Dates:	12/2025
Requested Funding:	\$1,780,740
Aim:	To investigate molecular recognition of phosphorylated biosubstrates
Status:	Scientific Review Group review completed: Council review pending (Impact score: 41, submitted just in time paperwork)

Teaching

Course Teaching.

Formal Courses **Bold**. All other courses are research-based.

Term	Prefix	Number	Title	Enrollment
25S	CHM	2210	Organic Chemistry I	175
	CHM	6935	Graduate Seminars in Chemistry	81
	CHM	7820	Directed Research	2
	CHM	7980	Dissertation: Doctoral	2
	CHM	4970	Undergraduate Research	1
24F	CHM	6263/4932	Advanced Organic Chemistry II: Physical Organic Chemistry	16
	CHM	6935	Graduate Seminars in Chemistry	74
	CHM	7820	Directed Research	2
	CHM	7980	Dissertation: Doctoral	2
24S	CHM	2210	Organic Chemistry I	175
	CHM	6935	Graduate Seminars in Chemistry	70
	CHM	7820	Directed Research	3
	CHM	7980	Dissertation: Doctoral	1
	CHM	4970	Undergraduate Research	1
23F	CHM	6263/4932	Advanced Organic Chemistry II: Physical Organic Chemistry	19
	CHM	7820	Directed Research	4
23S	CHM	2210	Organic Chemistry I	177
	CHM	7820	Directed Research	4
22F	CHM	6263/4932	Advanced Organic Chemistry II: Physical Organic Chemistry	13
	CHM	7820	Directed Research	2
	CHM	4970	Undergraduate Research	1

22S	CHM	7820	Directed Research	1
21F	CHM	6263/4932	Advanced Organic Chemistry II: Physical Organic Chemistry	16

Research Training Postdoctoral Associates

<i>Name</i>	<i>Dates</i>	<i>Representative Publication</i>
Canjia Zhai, Ph.D. US	June 2022– Present	ACS. Cent. Sci. 2025/ Chem. Eur. J. 2025 / Trends. Chem. 2025, 7, 70/ Chem. Sci, 2024, 15, 19588/ Chem. Sci. 2024, 15, 16040 / Chem. Eur. J. 2023, e202300524

Graduate Students

<i>Name</i>	<i>Degree</i>	<i>Date</i>	<i>Publication Record</i>
Chengkai Xu	PhD	Mar. 2022– Present	Chem. Sci, 2024, 15, 19588/ Chem. Sci. 2024, 15, 16040 / J. Mater. Chem. A. 2023, 11, 12214
Alexander Mariscal	PhD	Aug. 2022– Present	Chem. Eur. J. 2025 / Chem. Sci, 2024, 15, 19588 / Trends. Chem. 2025, 7, 70
Ipsita Maji	PhD	Aug. 2022– Present	N.A.

Undergraduate Students

Students in **bold** have published peer-reviewed papers

<i>Name</i>	<i>Date</i>	<i>Position After Graduation</i>
Briana Prieto	Aug.2024– May 2026	N.A.
Sebastian Mendez	Aug. 2024–May 2026	N.A.
Angus Rhoton	May 2025–Now	N.A.
Minh Hyynh	Jul 2026– Now	N.A.
Hannah Armanious	Apr. 2024–Present	medical school at FSU
Andrew Victoria	Jan. 2024–Present	N.A.
Thi Xuan Phan	Dec. 2023–May 2025	N.A.
Tobias Colmegna	Oct. 2023– May 2024	N.A.
Luzelena Sagal	Oct. 2023–May 2024	N.A.
Jay Conces	Nov. 2023–Present	N.A.
Sandra Mampilly	Nov. 2023–Present	N.A.
Haden Fisher	Oct. 2023–Present	N.A.
Ethan Zulueta	Mar. 2023–May 2024	Master student at USF
Quy Tran	Apr. 2022–May 2024	Sarasota Memorial Hospital.
Carson Doan	Jan. 2022–May 2023	West Virginia School of Osteopathic Medicine.
Thao Nguyen	Jan. 2022–Jan. 2023	N.A.
Alexander Mariscal	Jan. 2022–Jul 2022	Graduate student at USF

SERVICE

Service to the Department of Chemistry and USF

NSF REU students host	May 2025–July 2025
Specific Activities	• Advisor for REU students
NSF Career Awards Workshop Panelist	Nov. 15 2025
Specific Activities	• Participate in panel discussion • provide advice for early career scholars at USF
NMR committee	Aug. 2024–Present
Specific Activities	• Participate in committee discussion • Identify issues in NMR core facility and propose solutions to address them

Department Seminar Chair	Jan. 2024–May 2025
Specific Activities	• Coordinate seminar speaker visiting schedules • Invite name lecture speakers • Send out weekly emails for department seminars
Graduate Student Recruiting Committee	2021– 2024
Specific Activities	• Run departmental recruiting of new graduate students. • Run admissions process to select incoming graduate students

Member of Graduate Student Candidacy, Masters & Doctoral Committees

Name	Institutes /Department	Faculty Advisor	Year
Haiqiang Yang	USF / Pharmaceutical Sciences	Chuanhai Cao	2024–Present
Ruixuan Gao	USF / Chemistry	Jianfeng Cai	2022 – 2023
Chenhuan Wang	USF / Chemistry	Xiaodong Shi	2023
Menglin Xue	USF / Chemistry	Jianfeng Cai	2023
Alexandra Saputo	Moffitt / Drug Discovery	Justin Lopchuk	2021 – Present
Lubem Agbendeh	USF / Molecular Medicine	Xingmin Sun	2023 – Present
Michaela Senn	USF/ Chemistry	Randy Larsen	2023 – Present
Mina B. Sharabiani	USF /Chemistry	Ioannis Spanopoulos	2023 – Present
Stephanie Frost	USF / Chemistry	Jeffrey Raker	2022 – 2025
Xue Zhao	USF / Chemistry	Jianfeng Cai	2021 – Present
Yuyu Win	USF / Chemistry	Jianfeng Cai	2022 – Present
Jarais Fontaine	USF / Chemistry	Jianfeng Cai	2023– Present

Service to the community

St. Petersburg Science Festival	2025 –Present
Specific Activities	• Developing engaging science demonstrations based on our research • Engage in interactions with> 1500 audiences during the activities
R club:	2022– Present
Specific Activities	• Developing engaging science demonstrations based on our research • Visit 4 middle schools each year • Reach 120–150 middle schools students each year • Spark their interest in pursuing STEM degrees
USF Upward Bound	2023 – Present
Specific Activities	• Organize an annual summer camp • Accommodate 20–30 high school students in teaching labs • Hand-on lab training • Deliver career workshop for STEM fields

Service to the Profession

External Thesis Committee	05/13/2025 –05/15/2025
Specific Activities	External committee member for a PhD student at Dartmouth College
NSF Panel Reviewer	03/ 2025
Specific Activities	Reviewer for NSF grant applications
ACS PRF ad hoc Reviewer	06/ 2025
Specific Activities	Reviewer for ACS PRF grant applications
Tenure package reviewer	09/2024
Specific Activities	External tenure package reviewer for a PI at Huazhong University of Science and Technology
Journal Reviewer	2021– Present
Specific Activities	Reviewer for The Journal of American Chemical Society / JACS Au / Angewandte Chemie International Edition / Nature Communication / Chemical Science / Organic Letter / Macromolecules /ACS Sensors / Inorganic Chemistry / Communication Chemistry / Advanced Science / Analytical Chemistry / Beilstein Journal of Organic

TIRIS Grant Reviewer Specific Activities	2023 • Reviewer for TIRIS program from France • Evaluate postdoctoral fellowship applications
Castle Conference Judge Specific Activities	2022–Present • Poster judge • Oral presentation judge